



**SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR**

**Constituent of Symbiosis International (Deemed University), Pune**

(Established under Section 3 of the UGC Act of 1956 wide notification number F-9-12/2001-U-3 of Government of India)

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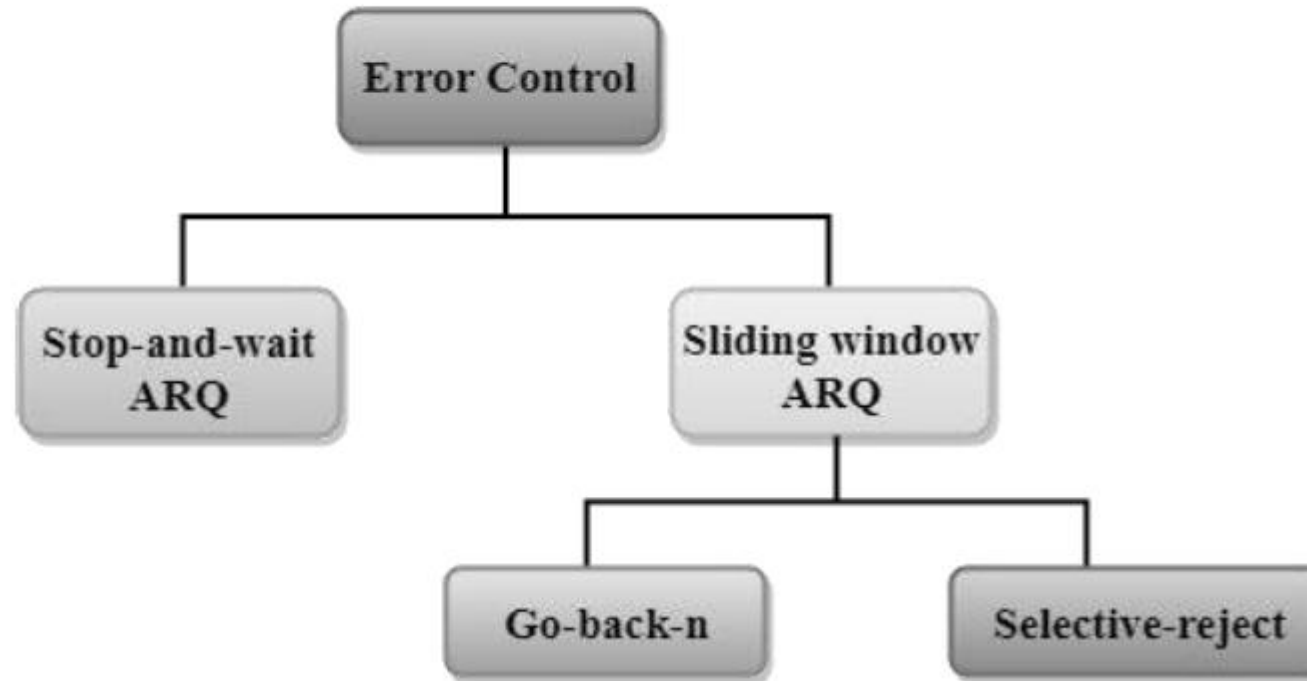


# Computer Networks/0705210503

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## Categories of Error Control



Error control in the data link layer is based on **automatic repeat request (ARQ)**.



## STOP-AND-WAIT ARQ

- Stop-and-wait ARQ is a technique used to retransmit the data in case of damaged or lost frames.
- This technique works on the principle that the sender will not transmit the next frame until it receives the acknowledgement of the last transmitted frame.

### Two possibilities of the retransmission in Stop and Wait ARQ:

- **Damaged Frame:** When the receiver receives a damaged frame(i.e., the frame contains an error), then it returns the NAK frame. For example, when the frame DATA 1 is sent, and then the receiver sends the ACK 0 frame means that the data 1 has arrived correctly. The sender transmits the next frame: DATA 0. It reaches undamaged, and the receiver returns ACK 1. The sender transmits the third frame: DATA 1. The receiver reports an error and returns the NAK frame. The sender retransmits the DATA 1 frame.

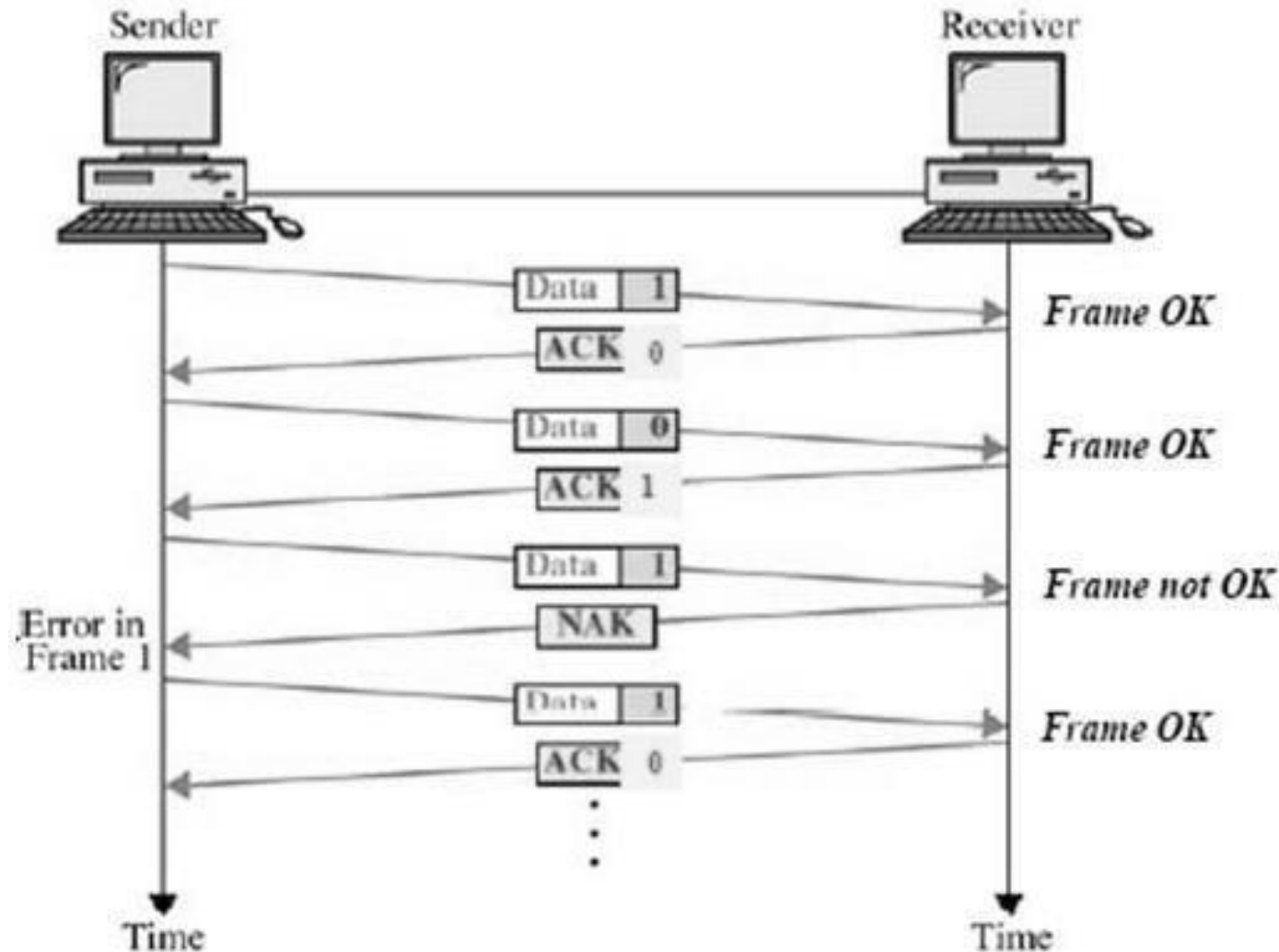


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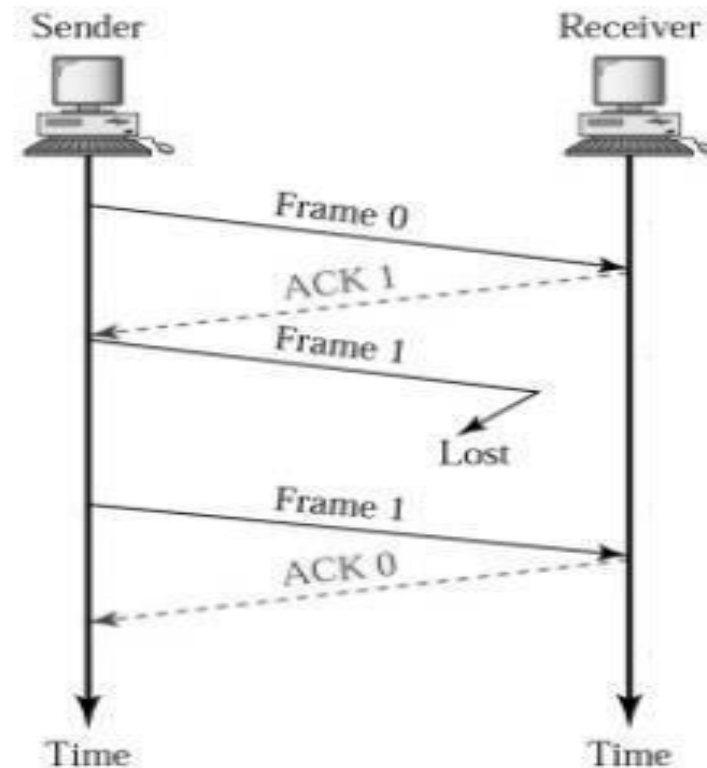
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**Lost Frame:** Sender is equipped with the timer and starts when the frame is transmitted. Sometimes the frame has not arrived at the receiving end so that it cannot be acknowledged either positively or negatively. The sender waits for acknowledgement until the timer goes off. If the timer goes off, it retransmits the last transmitted frame.





## SLIDING WINDOW ARQ

Sliding Window ARQ is a technique used for continuous transmission error control.

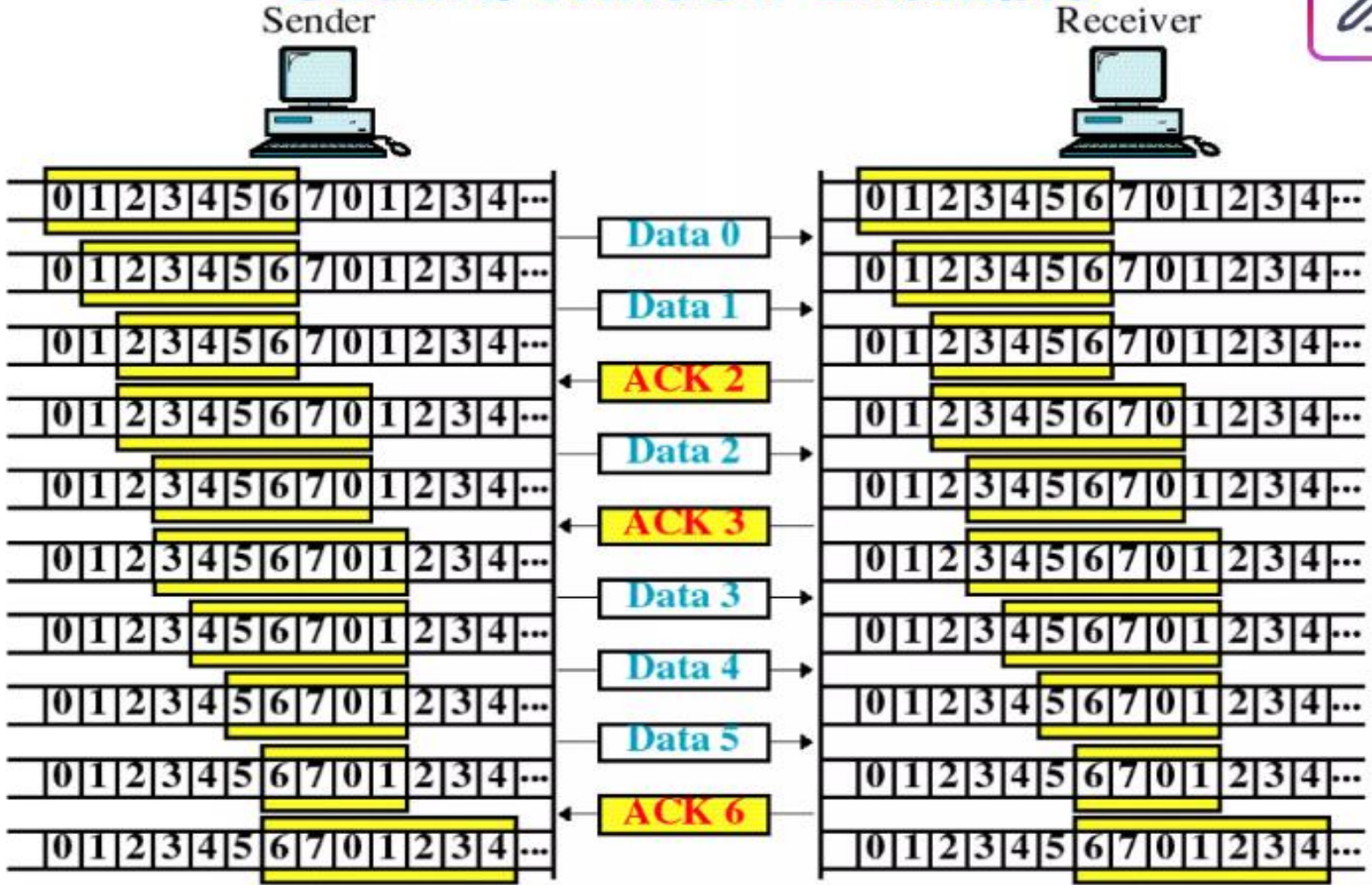
Two protocols used in sliding window ARQ:

### 1. GO-BACK-N ARQ

- In Go-Back-N ARQ protocol, if one frame is lost or damaged, then it retransmits all the frames after which it does not receive the positive ACK.



### Sliding Window Example





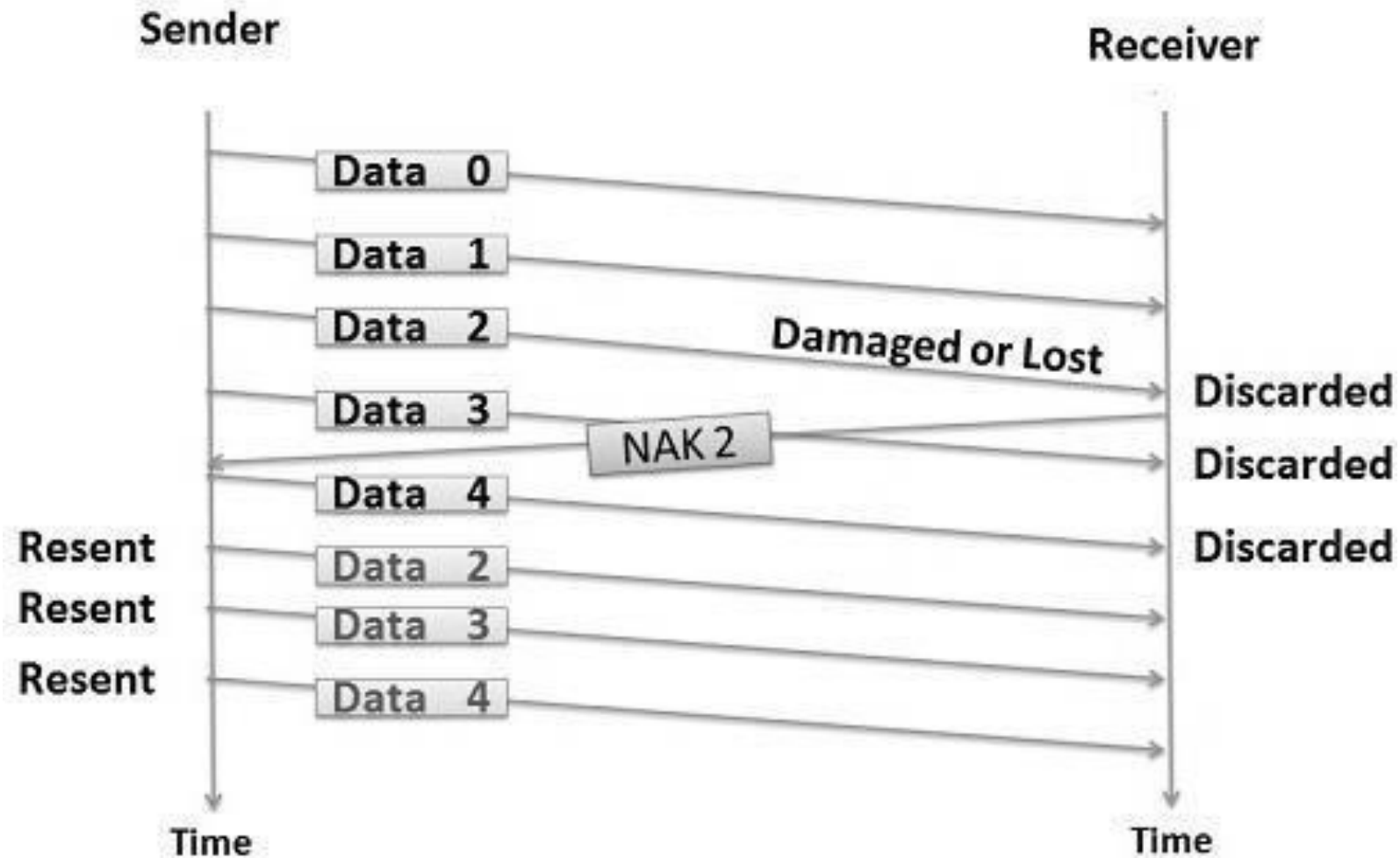
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- In the above figure, three frames (Data 0,1,2) have been transmitted before an error discovered in the third frame.
- The receiver discovers the error in Data 2 frame, so it returns the NAK 2 frame.
- All the frames including the damaged frame (Data 2,3,4) are discarded as it is transmitted after the damaged frame.
- Therefore, the sender retransmits the frames (Data2,3,4).

### **SELECTIVE-REJECT(REPEAT) ARQ**

- Selective-Reject ARQ technique is more efficient than Go-Back-n ARQ.
- In this technique, only those frames are retransmitted for which negative acknowledgement (NAK) has been received.
- The receiver storage buffer keeps all the damaged frames on hold until the frame in error is correctly received.
- The receiver must have an appropriate logic for reinserting the frames in a correct order.
- The sender must consist of a searching mechanism that selects only the requested frame for retransmission.

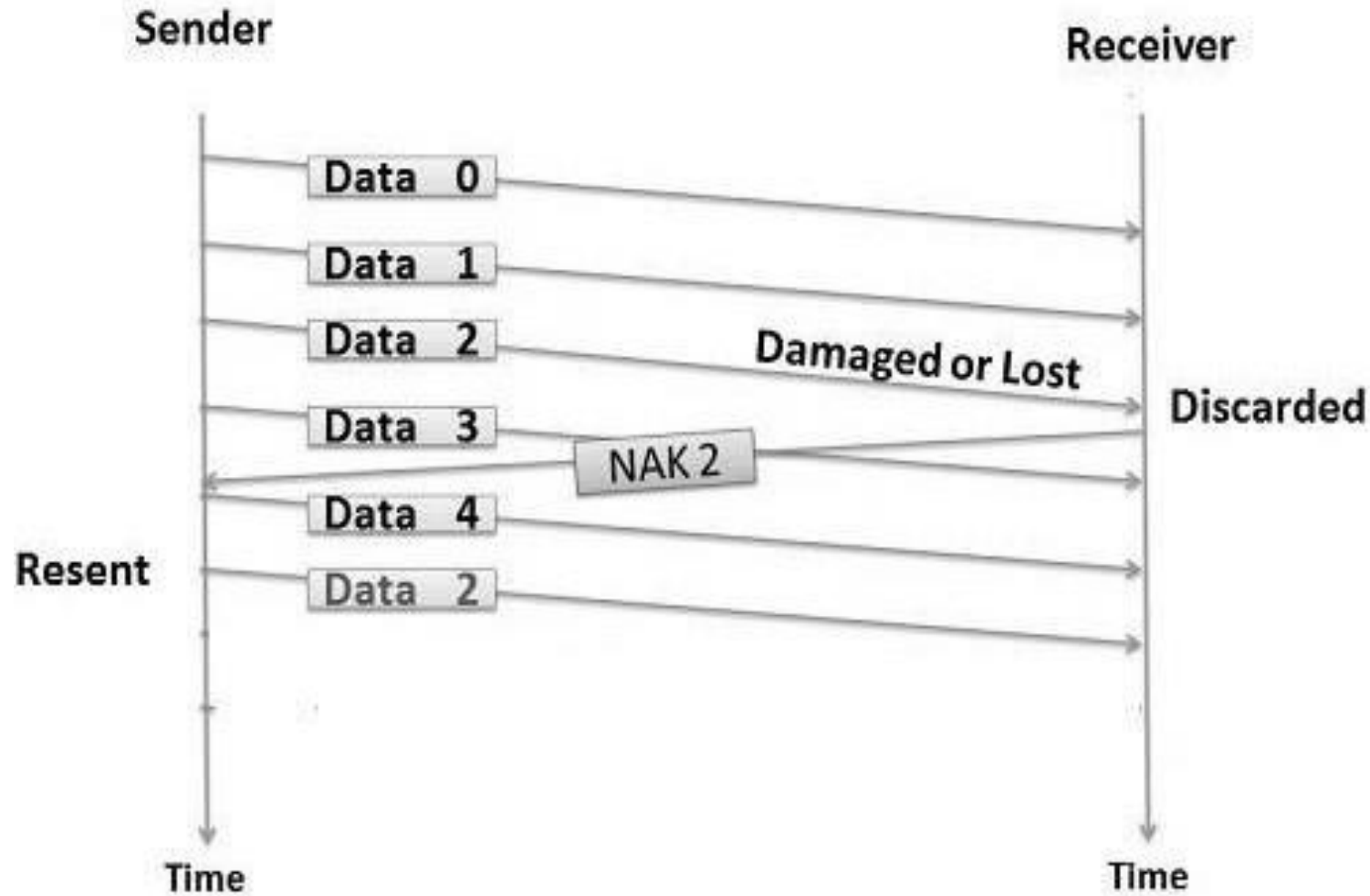


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- In the above figure, three frames (Data 0,1,2) have been transmitted before an error discovered in the third frame.
- The receiver discovers the error in Data 2 frame, so it returns the NAK 2 frame.
- The damaged frame only (Data 2) is discarded.
- The other subsequent frames (Data 3,4) are accepted.
- Therefore, the sender retransmits only the damaged frame (Data2).



### Problem:

Station A needs to send a message consisting of 9 packets to station B using a sliding window (window size 3) and go back n error control strategy. All packets are ready and immediately available for transmission.

If every 5th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

### Solution-

Given-

- Total number of packets to be sent = 9
- Go back N is used where  $N = 3$
- Every 5th packet gets lost



### Step-01:

Since sender window size is 3, so sender sends 3 packets (1, 2, 3)-



Total packets sent till now from sender side = 3

### Step-02:

After receiving the acknowledgement for packet-1, sender slides its window and sends packet-4.



Total packets sent till now from sender side = 4



### Step-03:

After receiving the acknowledgement for packet-2, sender slides its window and sends packet-5.



Total packets sent till now from sender side = 5

### Step-04:

After receiving the acknowledgement for packet-3, sender slides its window and sends packet-6.



Total packets sent till now from sender side = 6



### Step-05:

After receiving the acknowledgement for packet-4, sender slides its window and sends packet-7.



Total packets sent till now from sender side = 7

### Step-06:

- According to question, every 5th packet gets lost.
- So, packet-5 gets lost and when time out occurs, sender retransmits packet-5.
- In Go back N, all the following packets are also discarded by the receiver.
- So, packet-6 and packet-7 are discarded by the receiver and they are also retransmitted.
- Thus, the entire window is retransmitted.



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So, we have-

|   |   |   |
|---|---|---|
| 7 | 6 | 5 |
|---|---|---|

Total packets sent till now from sender side = 10

Now, the next 5th packet that will be lost will be packet-7. (6, 7, 5, 6, 7)

## Step-07:

After receiving the acknowledgement for packet-5, sender slides its window and sends packet-8.

|   |   |   |
|---|---|---|
| 8 | 7 | 6 |
|---|---|---|

Total packets sent till now from sender side = 11



### Step-08:

After receiving the acknowledgement for packet-6, sender slides its window and sends packet-9.



Total packets sent till now from sender side = 12

### Step-09:

- According to question, every 5th packet gets lost.
- So, packet-7 gets lost and when time out occurs, sender retransmits packet-7 and the following packets.
- Thus, the entire window is retransmitted.

So, we have-



Total packets sent till now from sender side = 15



Now, the next 5th packet that will be lost will be packet-9. (8, 9, 7, 8, 9)

### Step-10:

After receiving the acknowledgement for packet-7, sender slides its window.



Total packets sent till now from sender side = 15

### Step-11:

After receiving the acknowledgement for packet-8, sender slides its window.



Total packets sent till now from sender side = 15



### Step-12:

- According to question, every 5th packet gets lost.
- So, packet-9 gets lost and when time out occurs, sender retransmits packet-9.

So, we have-



Total packets sent till now from sender side = 16

Finally, all the 9 packets got transmitted which took total 16 number of transmissions.



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## **Problem:**

In Go back 4, if every 6th packet that is being transmitted is lost and if total number of packets to be sent is 10, then how many transmissions will be required?